CS 4412 M1 Project Proposal

Griffin Vines

Febuaray 5th, 2026

1 Dataset Description

My main datasets are the IMDb top 250 movies and Box Office Mojo's top 200 highest grossing movies, with Rotten Tomatoes acting as a bridge between them. Each of my sources represents the main points of my discovery questions; IMDb has the general response, BOM has the profits, and RT has the individual audience and critic responses. I plan to use the first two datasets to create a generalized list of movies and then cross reference each element with Rotten Tomatoes.

Total Movie List Draft					
Film Title	IMDb Rating	Box Office Mojo Rank	Total Box Office	Audience Rating	Critic Rating
The Dark Knight	#3	#19	534M	91%	89%
Forrest Gump	#19	#92	330M	81%	81%
The Shawshank Redemption	#1	N/A	25M	95%	95%
Transformers: Dark of the Moon	N/A	#59	800M	50%	61%

Then I will create a punnet square to sort the movies according to their audience and reviewer scores. This will follow the same catagorization used on Rotten Tomatoes, where any approval rating lower than 60% is considered a bad grade. However, RT also has a certification for films with a large number of reviews and a rating of 75% or higher. My plan is to start with a three-way 60/75 split and see how the data plays out. If the output for mediorce films is too low, then I will reformat the gradings to only split at 60%.

Movie Rating P-Square Draft			
	High Audience Scores	Moderate Audience Scores	Low Audience Scores
High Critical Scores	Pulp Fiction	Mank	Isthar
Moderate Critical Scores	Moana 2	Transformers	Star Wars Episode 8
Low Critical Scores	Aquaman	Battleship	Smurfs 2

Once the various metrics have been established, the real graph begins. There will be 4 separate bar charts, one for each punnet variant, which shows the profits of each film. The x-axis represents the general public's response towards each film in ascending order. The y-axis is how much each movie earned at the worldwide box office. This is further divided into three sections- a high gross rating of over a billion dollars, a low gross rating of less than 500 million dollars, and a medium gross rating between the two values.

2 Discovery Questions

The big questions driving my project are:
1. How does the critical response affect the box office of a movie?
2. How does the general response affect the box office of a movie?
3. Which section of the movie-watching audience has more power over the box office as a whole?
and to a lesser extent:
4. Can good reviews keep a movie alive in the public consensus?

I wanted to ask these questions because of my extended contact with the media discussion side of the internet, for better or worse. I have seen people claim that a highly profitable film with good critical reviews had bribed reviewers because it had low audience ratings, and I have seen those same people claim that a profitable film with low critical ratings has a cult audience blindly giving the media their money. This inspired me to create the three intial driving questions so I could build a foundation for these takes to be analyzed. If two movies have the same critic score, but the one that more audiences liked made more money, then that is the true sign of quality.

However, once I took a step back from my personal histroy I discovered a very interesting answer to be asked. There is a term known as "Vindicated by History", where a work that failed in the former present becomes heavily praised in the current present. I was reminded of this phrase after I looked through both datasets and found that praised movies are less modern than profitable movies.

So, I decided to perform a side study about what types of films are ranked low or high and looking at their legacies. Since this is dependent on the results of the data mining, it won't be finalized for a while.

3 Planned Techniques

I plan to use clustering and classification to mine through my datasets to help smooth down the rough data into my percise rankings. For the sorting of the films, I will use the clustering technique of k-means to do so. The ratings of all the films will be analyzed by my algorithm and assigned to the closest centroid based on the tomatoe-meter referenced before, once for the professional reviewers and again for the casual demographic. The same principle will be applies to the profit graphs using the divisons of the y-axis. However, the creation of the data sets will be dealt with classification through decision trees. The idea is to train the BoxOfficeMojo dataset to classify movies by economic success based on if they made a billion dollars or not, then take the films that made less than a billion and repeat the process again. This would create a training model for the IMDb data that cross-references the current film with what BOM says about it and predicts it accrodingly.

4 Sources

Github Link: https://github.com/PcktCalc-Kraftwrk/CS4412-Project

Dataset links

"Movies: TV Shows: Movie Trailers: Reviews." Rotten Tomatoes, www.rottentomatoes.com/.
"Top Lifetime Grosses." Box Office Mojo, www.boxofficemojo.com/chart/ww$_{t}op_{l}ifetime_{g}ross/?area = XWW.$
$\backslash IMDbTop250Movies.''IMDb, www.imdb.com/chart/top/.$